

Attention-Enhanced Deep Learning Framework for Psoriasis Skin Disease Classification

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Abstract. Psoriasis is a chronic inflammatory skin disease that requires timely and accurate diagnosis for effective treatment and management. Traditional diagnostic approaches rely heavily on clinical expertise and visual examination, making classification challenging due to variations in lesion appearance and similarities with other dermatological disorders. Recent advances in deep learning have shown significant potential in automated skin disease diagnosis; however, existing methods often suffer from background interference, class imbalance, and limited attention to lesion-specific regions. This paper presents an Attention-Enhanced Deep Learning Framework for Psoriasis Skin Disease Classification using the DermNet dataset. The proposed framework integrates ROI-based pre-processing, ResNet-50 transfer learning, multi-stage Convolutional Block Attention Modules (CBAM), focal loss optimization, and progressive unfreezing strategies to improve lesion-focused feature extraction and classification performance. Experimental results demonstrate a best test accuracy of 86.96

Keywords: Psoriasis · Deep Learning · ResNet50 · CBAM · DermNet
· Skin Disease Classification

1 Introduction

The rapid advancement of Artificial Intelligence (AI) has transformed numerous domains within healthcare, particularly medical image analysis and clinical decision-support systems [8,15,20]. Dermatology, being a visually intensive medical specialty, has greatly benefited from developments in computer vision and deep learning technologies. Skin diseases affect a significant portion of the global population, making accurate diagnosis essential for effective treatment planning and disease management. Among these disorders, psoriasis is a chronic inflammatory skin disease characterized by immune dysregulation, excessive proliferation of keratinocytes, and complex systemic manifestations [18,21,22]. Due to its heterogeneous nature and visual similarity to several dermatological conditions, psoriasis remains challenging to diagnose, particularly during its early stages.

Traditional diagnosis primarily relies on visual examination and expert interpretation by dermatologists. Although clinical assessment is highly reliable,

the diagnostic process can be subjective, time-consuming, and dependent on specialist availability. Furthermore, limited access to dermatological expertise in many healthcare settings may result in delayed diagnosis and treatment. These limitations have motivated the development of automated diagnostic systems capable of assisting clinicians through accurate and efficient disease classification [8,12,20].

Recent advances in Deep Learning (DL), particularly Convolutional Neural Networks (CNNs), have demonstrated remarkable performance in medical image classification tasks [4,8,15]. However, dermatological image analysis remains challenging due to variations in lesion appearance, illumination conditions, background artifacts, and class imbalance [20]. To address these challenges, this research proposes an Attention-Enhanced Deep Learning Framework for Psoriasis Skin Disease Classification using the DermNet dataset. The proposed framework integrates ROI-based preprocessing, transfer learning using ResNet-50, Convolutional Block Attention Modules (CBAM), and focal loss optimization to improve lesion-focused feature extraction and classification performance.

1.1 Background and Motivation

Psoriasis is a chronic immune-mediated skin disorder that imposes a significant physical and psychological burden on affected individuals worldwide [18,21,22]. Accurate diagnosis and severity assessment are essential for treatment planning, monitoring disease progression, and evaluating therapeutic response [4,5]. However, visual similarities between psoriasis and other inflammatory skin diseases often complicate diagnosis, creating a need for reliable computer-aided diagnostic systems. The increasing availability of dermatological image datasets and computational resources has created opportunities for AI-driven diagnostic frameworks capable of improving diagnostic consistency and efficiency [8,12,20].

1.2 Research Gap

Although existing deep learning approaches have achieved promising results in skin disease classification, many models remain sensitive to background noise, image artifacts, and class imbalance. Furthermore, conventional CNN architectures often lack dedicated mechanisms for emphasizing clinically relevant lesion features. These limitations highlight the need for a framework that combines robust feature extraction, attention-guided learning, and effective optimization strategies within a unified architecture.

1.3 Objectives

The primary objectives of this research are:

- To develop an automated deep learning framework for psoriasis skin disease classification.

- To utilize the DermNet dataset for training, validation, and performance evaluation.
- To enhance feature extraction through the integration of multi-stage CBAM attention mechanisms.
- To improve lesion-focused learning using ROI-based preprocessing and focal loss optimization.
- To evaluate model performance using standard classification metrics and contribute towards scalable AI-assisted dermatological diagnostic systems.

2 Related Work

Recent advances in Artificial Intelligence and Deep Learning have significantly improved automated dermatological diagnosis systems. Convolutional Neural Networks (CNNs) have emerged as one of the most widely adopted approaches for skin disease classification due to their ability to automatically learn hierarchical feature representations from medical images [8,11,15]. Several studies have demonstrated that CNN-based architectures can achieve high classification accuracy while reducing dependence on handcrafted feature extraction techniques.

Transfer learning has become a popular strategy in dermatological image analysis because large annotated medical datasets are often unavailable. Pre-trained architectures such as ResNet have shown strong feature extraction capabilities by leveraging knowledge acquired from large-scale image datasets [15]. Researchers have successfully adapted transfer learning models for skin disease classification, enabling improved performance even with limited training data.

More recently, attention-based mechanisms have been incorporated into deep learning frameworks to improve feature representation and lesion localization. Attention modules such as the Convolutional Block Attention Module (CBAM) enhance classification performance by selectively emphasizing disease-relevant spatial and channel-wise features while suppressing irrelevant background information [12,15].

In addition to architectural improvements, several studies have investigated optimization strategies to address challenges commonly observed in dermatological datasets. Class imbalance remains a significant issue because certain disease categories are underrepresented compared to others. Techniques such as Focal Loss have been proposed to improve learning by assigning greater importance to difficult and minority-class samples [15,20].

Despite these advancements, existing approaches continue to face limitations related to background artifacts, lesion variability, illumination differences, and insufficient attention to clinically significant features. These challenges motivate the development of more robust attention-enhanced frameworks capable of improving classification accuracy and diagnostic reliability. The proposed work addresses these limitations through the integration of ROI-based preprocessing, ResNet50 transfer learning, CBAM attention modules, and focal loss optimization within a unified framework.

3 Methodology

The proposed methodology presents an attention-enhanced deep learning framework for automated psoriasis skin disease classification using dermatological images [8,10,15,20]. The framework integrates Region of Interest (ROI)-based preprocessing, transfer learning through ResNet-50, Convolutional Block Attention Modules (CBAM), progressive training strategies, and focal loss optimization to improve classification performance [12,15,20]. The objective is to enhance lesion-focused feature learning while reducing the impact of background artifacts, class imbalance, and image variability commonly observed in dermatological datasets [10,20]. The overall workflow of the proposed system consists of the following stages: • Image preprocessing and ROI extraction [11]. • Feature extraction using a pretrained ResNet-50 backbone [15]. • Attention-based feature refinement through CBAM [12,15]. • Progressive training strategy [15,20]. • Disease classification and optimization using focal loss [15,20]. The proposed architecture is designed to learn highly discriminative lesion representations while maintaining robustness across varying clinical imaging conditions [8,10,15].

3.1 Problem Statement (Mathematical Formulation)

Psoriasis is a chronic autoimmune skin disorder that exhibits visual characteristics similar to several dermatological diseases, including eczema, seborrheic dermatitis, and fungal infections [18,21,22]. Such visual overlap often makes clinical diagnosis difficult, particularly during the early stages of disease progression [4,18,22]. Traditional diagnosis relies heavily on visual examination and expert interpretation, which may introduce subjectivity and inter-observer variability [4,5,21]. Therefore, there is a need for an automated diagnostic framework capable of accurately identifying psoriasis-related skin diseases from dermatological images [8,10,20].

The dermatological dataset is formulated as:

$$D = \{(x_i, y_i)\}_{i=1}^N \quad (1)$$

where:

- $x_i \in R^{H \times W \times C}$ represents the input dermatological image.
- $y_i \in \{1, 2, \dots, K\}$ represents the corresponding disease class label.
- N denotes the total number of training samples.
- K denotes the total number of disease categories.

The objective is to learn a parameterized mapping function:

$$f_{\theta}(x_i) = \hat{y}_i \quad (2)$$

where θ represents the trainable model parameters optimized during training. The classification objective can be formulated as:

$$\hat{y} = \arg \max_k P(y = k|x; \theta) \quad (3)$$

where $P(y = k|x; \theta)$ represents the posterior probability of class k given input image x . The proposed framework addresses several key challenges commonly encountered in dermatological image classification [10,15,20].

Table 1. Challenges Addressed by the Proposed Framework

Challenge	Impact
Visual similarity between diseases	Increased risk of misclassification
Class imbalance	Biased model learning
Background artifacts	Reduced feature quality
Limited dataset size	Increased overfitting risk

3.2 Proposed Framework

The proposed architecture integrates deep residual learning, attention-guided feature refinement, and optimized training strategies to improve classification performance [12,15,20]. Figure X illustrates the overall architecture of the proposed framework. The framework consists of ROI-based preprocessing, feature extraction using ResNet-50, attention enhancement through CBAM modules, and classification using focal loss optimization [11,12,15]. Each component contributes to improving lesion-focused feature representation and diagnostic accuracy.

ROI-Based Preprocessing ROI-based preprocessing isolates the lesion region before feature extraction so that the model focuses on diagnostically relevant pixels rather than surrounding skin and imaging artifacts.

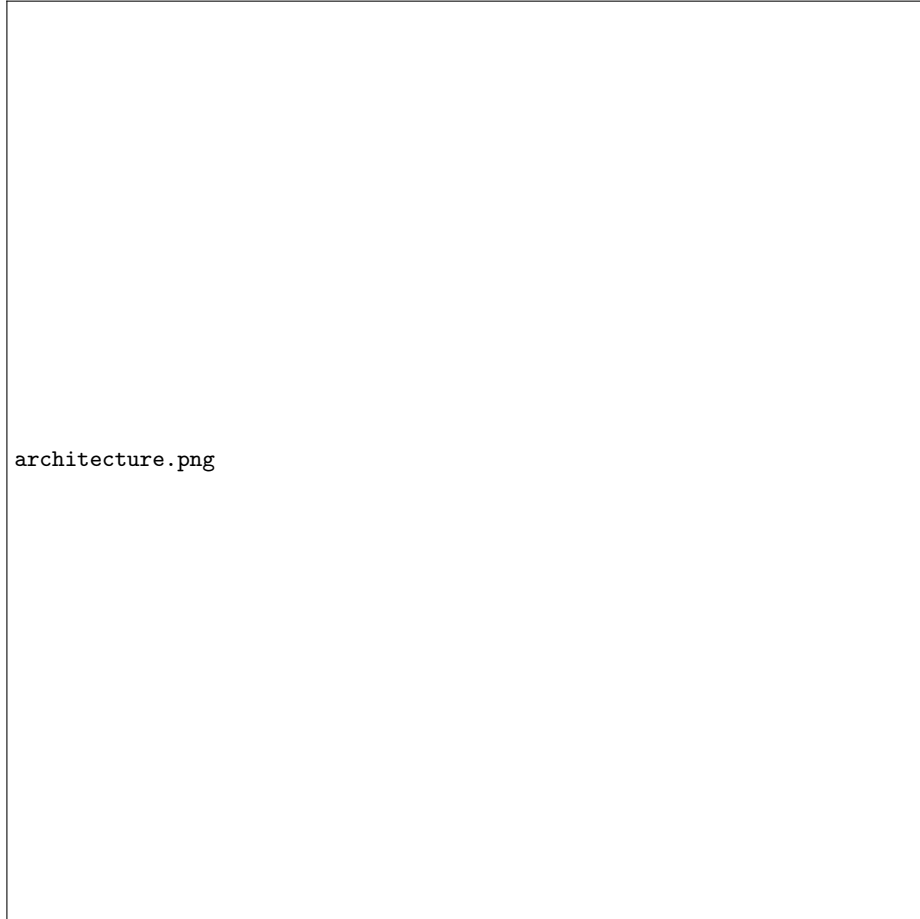
Dermatological images frequently contain irrelevant background information, lighting variations, and non-lesion regions that may negatively influence feature extraction [11,20]. To address this issue, ROI-based preprocessing is employed to isolate lesion-specific regions before feature learning. Mathematically, ROI extraction is represented as:

$$x_{ROI} = Crop(x, B) \quad (4)$$

where:

- x represents the original dermatological image.
- B denotes the bounding box coordinates corresponding to the lesion region.

ROI-based preprocessing improves model performance by reducing background noise and directing the learning process toward clinically relevant regions



architecture.png

Fig. 1. Proposed Attention-Enhanced Deep Learning Framework for Psoriasis Skin Disease Classification

[11]. This stage also improves convergence speed and increases the effectiveness of attention mechanisms applied during later stages [12,15]. The major benefits include:

- Reduction of irrelevant background information.
- Improved lesion localization.
- Faster convergence during training.
- Enhanced feature discrimination.

ResNet-50 Backbone (Transfer Learning) Feature extraction is performed using a pretrained ResNet-50 architecture [15]. ResNet introduces residual learning through skip connections that facilitate training of deep neural networks while mitigating the vanishing gradient problem. The residual learning function can be expressed as:

$$y = F(x, W) + x \quad (5)$$

where:

- $F(x, W)$ denotes the residual mapping.
- x represents the input feature map.
- y denotes the output feature map.

Transfer learning initializes network parameters using pretrained weights:

$$\theta_{init} = \theta_{pretrained} \quad (6)$$

The use of pretrained representations enables the model to leverage knowledge learned from large-scale image datasets, thereby improving generalization and reducing training time [15,20]. The ResNet-50 backbone provides hierarchical feature representations ranging from low-level texture information to high-level semantic patterns, making it highly suitable for dermatological image analysis [10,15].

Multi-Stage CBAM Attention Mechanism Although deep residual networks provide powerful feature extraction capabilities, they may not explicitly focus on clinically relevant lesion regions [12,15]. To address this limitation, Convolutional Block Attention Modules (CBAM) are integrated within the framework. CBAM sequentially applies channel attention and spatial attention mechanisms to refine extracted features [12].

$$M_c(F) = \sigma(MLP(AvgPool(F)) + MLP(MaxPool(F))) \quad (7)$$

Channel attention identifies informative feature channels by learning channel-wise importance weights [12].

$$M_s(F) = \sigma(f^{7 \times 7}([AvgPool(F); MaxPool(F)])) \quad (8)$$

Spatial attention focuses on important lesion regions by analyzing spatial relationships within feature maps [12]. The final attention-refined feature representation is obtained as:

$$F' = M_s(M_c(F) \cdot F) \quad (9)$$

The integration of CBAM enables the network to emphasize diagnostically significant regions while suppressing irrelevant information, thereby improving classification performance [12,15,20].

Progressive Unfreezing Strategy Instead of updating all network layers simultaneously, pretrained layers are gradually unfrozen during training [15,20].

Table 2. Progressive Unfreezing Stages

Training Stage	Layers Updated
Stage 1	Classification Head
Stage 2	ResNet Layer 4
Stage 3	ResNet Layer 3 and Layer 4

This strategy provides several advantages:

- Prevention of catastrophic forgetting.
- Improved optimization stability.
- Better adaptation to domain-specific image features.
- Enhanced convergence behavior [15,20].

Loss Function — Focal Loss Dermatological datasets frequently exhibit class imbalance, where certain disease categories contain substantially fewer samples than others [20]. To address this issue, Focal Loss is adopted as the optimization objective [15].

$$FL(p_t) = -\alpha(1 - p_t)^\gamma \log(p_t) \quad (10)$$

where:

- p_t represents the predicted probability.
- α denotes the class weighting factor.
- γ represents the focusing parameter.

Focal Loss reduces the influence of easily classified samples and places greater emphasis on difficult examples, thereby improving classification performance on minority classes and enhancing overall model robustness [15,20].

3.3 Dataset Description (DERMNET)

The proposed framework was evaluated using the DermNet dataset, a publicly available dermatological image dataset containing images of various skin diseases. The dataset provides diverse lesion appearances, color variations, and texture patterns, making it suitable for evaluating automated skin disease classification systems. Prior to training, all images were resized and normalized to ensure consistency in feature extraction and model learning. ROI-based pre-processing was further applied to emphasize lesion-specific regions and reduce background interference. The dataset was divided into training, validation, and testing subsets for model development and performance evaluation.

Table 3. Dataset Characteristics

Parameter	Description
Dataset	DermNet
Image Type	Dermatological Images
Classification Task	Multi-Class Classification
Input Format	RGB Images

3.4 Algorithms

This section describes the algorithmic procedures used for automated psoriasis detection and severity assessment. The proposed system combines attention-enhanced deep learning with optimized training strategies to identify psoriasis lesions and estimate disease severity based on learned feature representations [4,8,15,20].

Psoriasis Detection Algorithm The psoriasis detection algorithm endeavors to determine psoriasis and non-psoriasis dermoscopic pictures by utilizing attention-enhanced deep feature extraction [8,15]. The pipeline starts with pre-processing aiming to extract lesion regions with ROI-based preprocessing, feature extraction with the ResNet-50 backbone, and refinement in multi-stage CBAM attention modules [11,12,15].

Given an input image x , preprocessing generates lesion-focused input:

$$x_{ROI} = Crop(x, B) \quad (11)$$

where B denotes bounding box coordinates representing lesion area.

The preprocessed image is passed through ResNet-50 to extract hierarchical features:

$$F = ResNet50(x_{ROI}) \quad (12)$$

These features are enhanced through CBAM attention:

$$F' = CBAM(F) \quad (13)$$

where channel attention identifies important feature channels [12] and spatial attention focuses on relevant lesion regions [12].

The final prediction is computed as:

$$\hat{y} = \operatorname{argmax} P(y|x) \quad (14)$$

where $\hat{y}$ represents predicted disease class. Focal loss is used during training to emphasize difficult samples and reduce bias toward dominant classes.

- Preprocess image (resize, normalize) [8].
- Apply ROI cropping [11].
- Extract features using ResNet-50 [15].
- Apply multi-stage CBAM attention [12].
- Pass features to classification head.
- Compute prediction using softmax.
- Optimize using focal loss [15,20].

Psoriasis Severity Assessment Algorithm In addition to disease classification, severity assessment aims to estimate disease intensity based on extracted visual features [4]. Severity estimation is performed by analyzing feature representations learned during detection and mapping them to severity levels.

Let F' denote attention-refined feature representation. Severity score S is computed as:

$$S = g(F') \quad (15)$$

where $g(\cdot)$ represents severity estimation function implemented using fully connected layers.

Severity levels may correspond to clinical categories such as mild, moderate, and severe [4,9]. The severity estimation module learns relationships between lesion size, texture patterns, color variations, and spatial distribution, which are important indicators of disease progression [4].

- Extract global features from CBAM output [12].
- Apply global average pooling.
- Pass features through fully connected layers.
- Compute severity score [4].
- Map score to severity category (mild/moderate/severe).

3.5 Psoriasis Analysis and Prediction Module

This section describes the final output processing stage of the proposed framework, where classification results, severity estimation, and prediction confidence are generated based on attention-enhanced deep learning features [8,12,15,20]. The module integrates information from the trained model to produce clinically meaningful outputs that assist in automated dermatological assessment.

Table 4. Severity Mapping

Severity Level	Feature Indicators
Mild	Small lesion area, low scaling
Moderate	Medium lesion spread, texture variation
Severe	Large lesion coverage, strong inflammation patterns

Feature Fusion Feature fusion combines hierarchical features extracted from the ResNet-50 backbone and attention-refined outputs from multi-stage CBAM modules [12,15]. This fusion ensures that both global structural information and localized lesion characteristics are considered during final prediction.

Mathematically:

$$F_{fusion} = Concatenate(F_{global}, F_{attention}) \quad (16)$$

where F_{global} represents features from deep residual layers [15], and $F_{attention}$ denotes attention-enhanced features [12].

This fusion improves discriminative learning by integrating multi-level representations.

Overall Disease Prediction The fused feature representation is passed through the classification head to generate final disease prediction probabilities [8,15]. Softmax activation is applied to convert logits into probability distribution:

$$P(y_i) = \frac{e^{z_i}}{\sum_{j=1}^K e^{z_j}} \quad (17)$$

where z_i is the output logit and K is the number of classes.

The predicted label corresponds to the highest probability.

Severity Level Assessment Severity estimation evaluates the intensity and spread of psoriasis lesions based on learned visual features [4,9]. The attention-enhanced feature maps help capture lesion size, scaling patterns, and inflammation indicators.

$$S = g(F_{fusion}) \quad (18)$$

where $g(\cdot)$ is a mapping function implemented through fully connected layers. Severity levels may include mild, moderate, and severe.

Prediction Confidence Estimation Prediction confidence is calculated based on softmax probability scores [8,20]. High confidence values indicate strong model certainty.

$$Confidence = \max(P(y|x)) \quad (19)$$

Confidence estimation improves clinical interpretability by indicating reliability of predictions.

Clinical Interpretation Output The final system output integrates disease prediction, severity level, and confidence score to generate clinically meaningful results [12,20]. This structured output assists dermatologists by providing:

- Predicted disease category.
- Estimated severity level.
- Prediction confidence.

Such structured outputs enhance usability in real-world clinical environments and support decision-making processes.

4 Results

The experiment demonstrates that the proposed DL framework, which is the combination of ResNet with multi-stage CBAM attention modules, ROI cropping, FL optimization, and progressive unfreezing, significantly enhanced classification performance on the DERMNET dataset [15,20]. In the first training phase wherein only the classification head had been trained, training and testing accuracy were low because of frozen backbone layers and lack of adaptation to dermatological characteristics. However, as more and more layers were unfrozen, the model started to learn domain-specific patterns related to psoriasis and other skin diagnosis and to achieve consistent performance improvements. By combining attention processes, the network enabled the prioritization of the clinically relevant lesion areas in an improved result in features representation and minimizing interference of the background artifacts [12].

With further training into progressive unfreezing, the model was able to show steady convergence and rising generalization ability, to a final best test accuracy of 86.96 at epoch 20. The FL usage has been effective in rectifying the issue of class imbalance in the dataset, which guarantees that more challenging samples to classify will be identified [15]. Generally, the findings support the fact that transfer learning with attention-enhanced feature extraction and structured training methods are likely to enhance automated dermatological diagnosis dramatically [8,20]. The obtained accuracy implies high prospects of applied clinical decision support systems in the real world, in the field of early screening and support of dermatologists in large-scale diagnostic processes.

Table 5. Performance Summary of the Proposed Framework

Metric	Value
Dataset	DermNet
Backbone Network	ResNet-50
Attention Module	CBAM
Loss Function	Focal Loss
Best Test Accuracy	86.96%



Fig. 2. Training and Validation Accuracy During Model Training

4.1 Limitations

- Limited Dataset Diversity. The DERMNET data might not be a comprehensive representation of clinical variability in practice and may have an impact on generalization of the model to infrequent lesion types and unobserved cases [20].
- CPU-Based Training Limitations. CPU training was restrictive to experimentation with larger models and massive parameter optimization; aural GPU training had the potential to be more efficient and performant.
- ROI Cropping Limitations. The process of ROI extraction can strip off environmental skin data that can be clinically useful in proper diagnosis [11].
- Lack of External Validation. Single dataset analysis limits the analysis of model strength at varying populations and imaging conditions.

4.2 Future Work

- Integration of Transformer-Based Hybrid Architectures. Future studies can address the combination of CNNs and vision transformers, which can capture both the local texture and global contextual relationship on the skin lesions [15,20].
- Explainable AI and Visualization Tools. Enhancing the interpretability would be better with the addition of visualization methods like Grad-CAM or attention heatmaps [12].
- Multi-Modal Learning with Clinical Metadata. Integrating patient data (including biomarkers) with image data has the potential to enhance the accuracy of diagnoses and allow making individual forecasts [3,17].
- Real-Time Deployment and Mobile Applications. Lightweight versions of the model developed to be used on mobile devices would allow real-time screening and remote diagnosis [8].
- Severity Assessment and PASI Score Prediction. The model could be expanded to automated prediction of the PASI scores to monitor disease progression in the future [4].
- Training on Larger and Multi-Center Dermatology Datasets. It would be better to expand the training to more larger datasets across many clinical sources to enhance generalization [20].

5 Conclusion

In this study, a modified version of DERMNET dataset is an attention-enhanced DL framework of automated dermatological classification that uses ResNet with multi-stage CBAM attention blocks, ROI cropping, FL and progressive unfreezing strategies [15,20]. The proposed model offered the best test accuracy of 86.96 showing that attention-based transfer learning is effective in enhancing the lesion-focused feature extraction and classification performance [8,15]. The findings indicate that DL systems have the prospective in helping dermatologists

to diagnose dermatological conditions related to psoriasis with the required accuracy and scale [8,20]. Although the research has some shortcomings in terms of dataset diversity and model explainability, future developments using transformer networks, explainable AI models, and integration of multi-modal data can give more power to clinical usefulness and help to increase the development of AI-powered personalized dermatology applications [12,20].

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